

BERT'S

Sheet Metal Products Inc.

Laser Fume Extraction Fan

The blue centrifugal blower beside the fiber laser. It pulls smoke and dust out of the cutting-table plenum through the flex hose and pushes it up the outlet duct.

Bert's Sheet Metal Products, Inc. · 9521 Irondale Ave, Chatsworth, CA 91311 · 818.775.0104 · BSMP-MG-003 Rev A

Nameplate

Values marked ~ are hard to read on the plate (worn and dirty). Correct them here when someone reads them in good light.

Field	On the plate	In US units / meaning
Maker	Dongguan Huayuan Ventilation Equipment Co., Ltd. (Dongguan, Guangdong, China)	
Product	Low Noise Centrifugal Ventilator	
Model (Type)	6-46-3A (QC sticker: 6-46 No.3A)	See "What the model number means" below
Power	3 kW	about 4 HP
Speed	~3455 r/min	2-pole motor running on 60 Hz
Flow	~2891 m³/h	about 1,700 CFM
Total pressure	~1452 Pa	about 5.8 in. water column
Voltage	380 V, 3-phase	Same supply as the laser and chiller (step-up transformer). Expect about 6–7 A per leg at full load.
Serial no.	~1051	
Built	~2024/01	Arrived with the laser (Bescutter, mid-2024)

- What the model number means
- **6-46** — the fan family (Chinese standard naming, like the common 4-72 and 9-19 series). 6-46 is a medium/high-pressure design: it trades some volume for suction, which is what you need to pull through a long hose and a filter.
 - **No.3** — impeller size number. No.3 = about **300 mm (12 in.)** impeller diameter.
 - **A** — drive type A = **direct drive**. The impeller sits on the motor shaft. No belt, no pulleys, no separate fan bearings. The only wear parts are the motor's own two bearings.



Maintenance schedule

How often	Task	Who
Daily	Listen when the laser starts: steady hum, no rattle or scraping. Check the hose is on and not crushed or kinked.	Laser operator
Weekly	Blow dust off the motor fins and the fan cover grille (power off). A dusty motor runs hot. Check the hose clamps and the tape on the outlet joint.	Laser operator
Monthly	Look for dust leaking out at the outlet flange and the scroll seams. Fresh dust on the housing = a leak; reseal with foil HVAC tape or gasket.	Laser operator
Every 6 months	Lock out, open the inlet and look at the impeller blades. Scrape off dust/dross buildup evenly (a lopsided cake throws it out of balance). Spin by hand: it should turn free and quiet.	Manager
Yearly	Electrician: amps per leg at full load (should be even, under the motor's FLA), terminal box tight and dry, insulation to ground.	Electrician
When needed	Smoke hanging over the table: check the filter downstream first, then hose and joints, then the fan.	Anyone

Safety

Always lock out the fan's power before reaching into the inlet or outlet. A 300 mm impeller at 3,450 rpm does not stop fast, and on a direct-drive fan the impeller IS the motor shaft.

Troubleshooting

Symptom	Most likely cause	What to do
Weak suction, smoke at the table	Loaded filter downstream, crushed/kinked hose, leaking joint, dampers closed	Filter first. Then walk the hose and the outlet duct for leaks. Then the fan.
Weak suction but the fan sounds normal	Running backwards. A centrifugal fan running in reverse still blows, just much weaker.	Look at the motor fan through the cover grille at start-up; match the arrow on the scroll. Wrong = swap any two of the three motor leads (electrician).
Vibration or new noise	Dust caked unevenly on the impeller; loose mounting bolts; motor bearing going	Lock out, clean the impeller, tighten the base bolts. Grinding or growling that stays = bearings.
Trips the overload / breaker	Lost phase, low voltage from the transformer, bearings dragging, impeller rubbing	Electrician: voltage and amps on all three legs. Spin by hand with power off.
Motor too hot to touch	Dirty fins, blocked fan cover, one leg low	Clean the fins and grille; have the amps checked.

Spares and replacement

- **Motor:** any 3 kW (4 HP), 2-pole, 380 V 3-phase, foot-mount (B3) or foot-and-flange (B35) motor. Read the frame size (probably 100L) and the shaft diameter off the motor's own label (round sticker on the fan cover) before ordering, and the impeller bore must match the shaft.
- **Motor bearings:** two deep-groove ball bearings, sizes printed on the motor label or stamped on the bearings (frame 100L is usually 6206 both ends).
- **Whole fan:** search "6-46-3A centrifugal fan 3kW" (Alibaba / Huayuan), or a US high-pressure blower rated about **1,700 CFM at 6 in. w.c.** with matching inlet/outlet sizes.

Fill in when known

Item	Value
Motor label: maker / frame / FLA / shaft	
Motor bearing numbers (front / back)	
Inlet hose size / outlet duct size	
Filter downstream: make / element part no.	
Amps per leg at full load (date)	